

Advanced economics

Chp:- Demand and Supply

Topic:- Calculation of PED

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Calculation of Price Elasticity of demand

Formula:-

Price Elasticity of demand :- $\frac{\text{percentage change in quantity demand}}{\text{percentage change in price}}$

% Δ (percentage change) :- $\frac{\text{New} - \text{old}}{\text{old}} \times 100$

percentage change in quantity demand : $\frac{\text{New } Q - \text{old } Q}{\text{old } Q} \times 100$
(% Δ in QD)

percentage change in price : $\frac{\text{New } P - \text{old } P}{\text{old } P} \times 100$
(% Δ in Price)

Case #1 :-old
New

-ve	
Price	QD
↑ 10	↓ 100
20	50

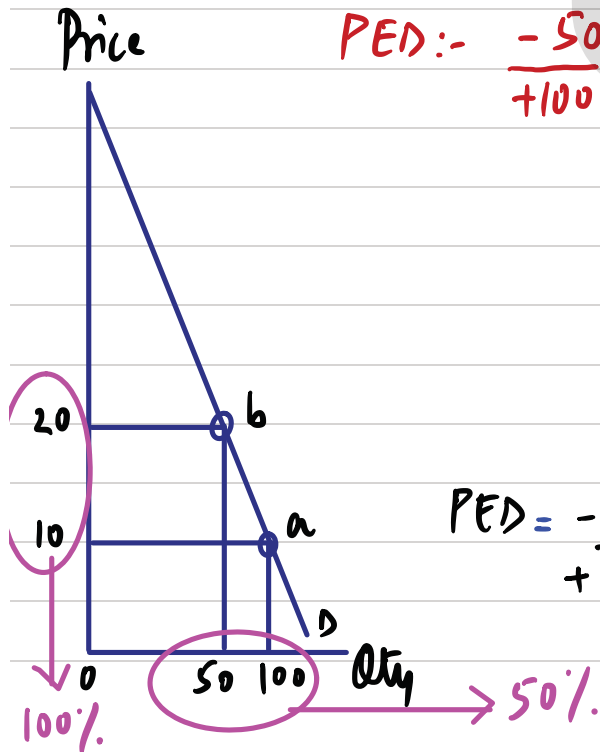
PED: ?

$$\% \Delta \text{ in QD} :- \frac{\text{New Q} - \text{old Q}}{\text{old Q}} \times 100 :- \frac{50 - 100}{100} \times 100 \Rightarrow -50$$

$$\% \Delta \text{ in Price} :- \frac{\text{New P} - \text{old P}}{\text{old P}} \times 100 :- \frac{20 - 10}{10} \times 100 \Rightarrow +100$$

$$\text{PED} :- \frac{-50}{+100} \Rightarrow -0.5 (< 1) \text{ Inelastic}$$

This sign represents negative relationship between price and quantity demand which is always negative, that price and quantity demand goes in opposite direction. So always ignore negative sign while calculating PED!



$$\text{PED} = \frac{-50\%}{+100\%} \Rightarrow -0.5$$

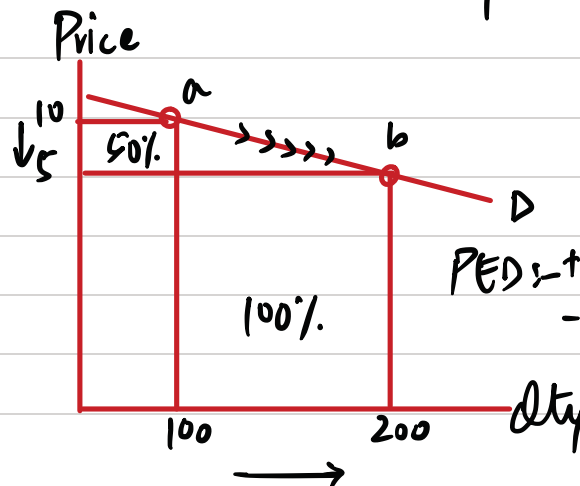
Case #2old
new

Price	QD
↓ 10 5	↑ 100 200

$$\% \Delta \text{ in } QD \Rightarrow \frac{\text{New } Q - \text{old } Q}{\text{old } Q} \times 100 \Rightarrow \frac{200 - 100}{100} \times 100 \Rightarrow +100\%$$

$$\% \Delta \text{ in price} \Rightarrow \frac{\text{New } P - \text{old } P}{\text{old } P} \times 100 \Rightarrow \frac{5 - 10}{10} \times 100 \Rightarrow -50\%$$

$$PED:- \frac{\% \Delta \text{ in } QD}{\% \Delta \text{ in price}} \Rightarrow \frac{+100\%}{-50\%} \Rightarrow -2 (>1 \text{ Elastic})$$



$$PED = \frac{+100\%}{-50\%} = -2 \text{ (Elastic demand)}$$

This sign represents negative relationship between price and quantity demand. Always ignore this sign to identify the elasticity, else every answer will be considered inelastic by you!

case #3old
new

Price	QD
10	100
15	100

→ PED ?

→ PED = 0

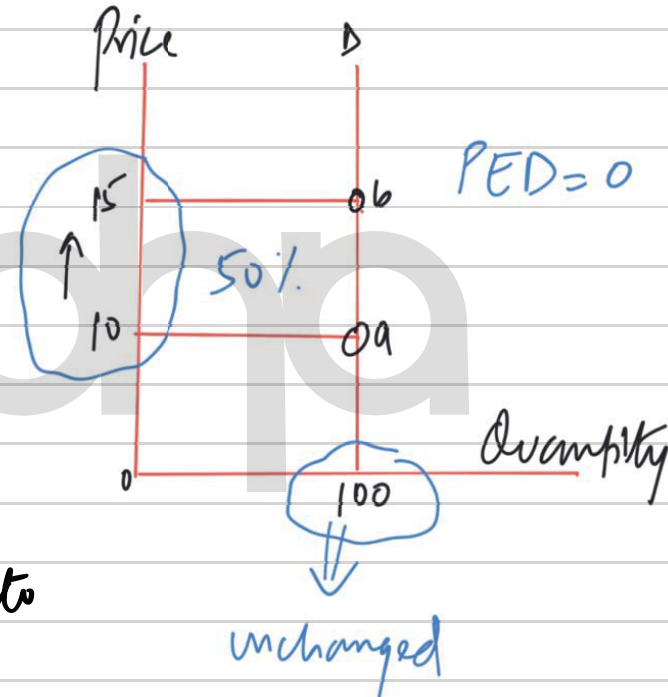
Perfectly inelastic demand

$$\% \Delta := \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

$$\% \Delta \text{ in QD} := \frac{100 - 100}{100} \times 100 = 0$$

$$\% \Delta \text{ in price} := \frac{15 - 10}{10} \times 100 = 50\%$$

$$\text{PED} := \frac{0}{50\%} \Rightarrow 0 \quad (\text{PED} = 0)$$



* Zero divided by anything is zero

* Quantity unchanged was a clear hint to identify the elasticity of demand

case #4

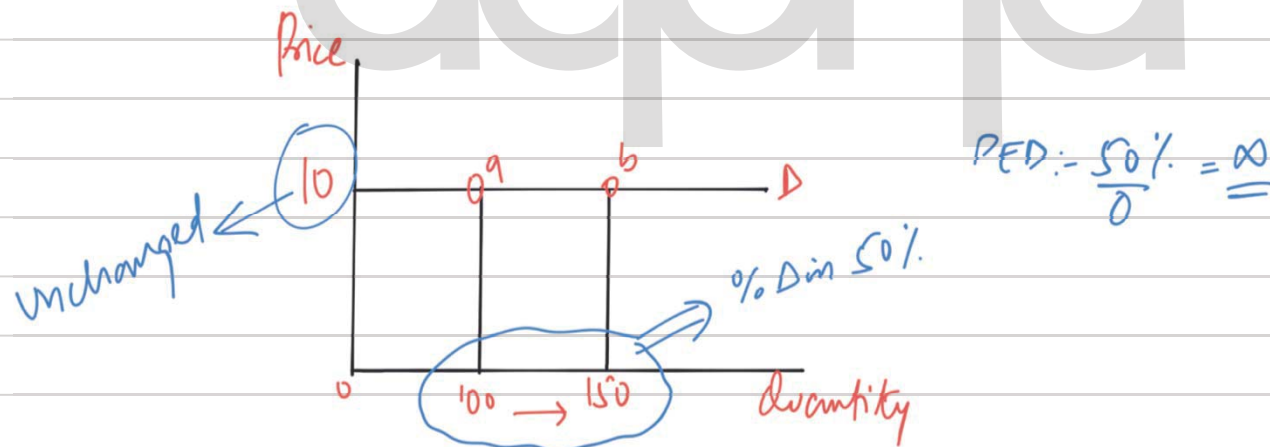
	price	QD
old	10	100
new	10	150

PED = ? % Δ := $\frac{\text{New} - \text{old}}{\text{old}} \times 100$

% Δ in QD := $\frac{150 - 100}{100} \times 100 = 50\%$

% Δ in Price := $\frac{10 - 10}{10} \times 100 = 0$

PED := $\frac{50}{0} \Rightarrow$ Math error, infinite ∞ , PED = ∞



* Anything divided by zero will result in infinity

* Unchanged price level was a clear hint to identify the elasticity of demand.

case #s :-

Old
New

Price	QD
↓ 10	↑ 100
5	150

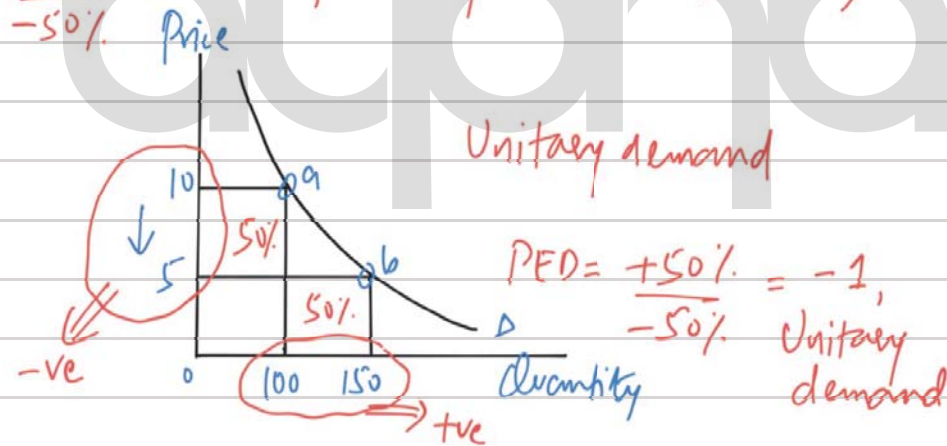
PED:-?

$$\% \Delta := \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

$$\% \Delta \text{ in QD} := \frac{150 - 100}{100} \times 100 = +50\%$$

$$\% \Delta \text{ in price} := \frac{5 - 10}{10} \times 100 = -50\%$$

$$\text{PED} := \frac{+50\%}{-50\%} = -1, \text{ Unitary demand (PED} = -1)$$



Unitary demand, a change in price brings an equal percentage change in quantity demand. For e.g. a rise in price by 50% will contract quantity demand by 50%, keeping total revenue or total expenditure unchanged.

Question

Marks:- 14

Q1 Using a diagram and a numerical example, explain the importance of negative sign under price elasticity of demand? 6

Ans:- * Take numerical e.g. $\Rightarrow$ calculation of elastic demand
Diagram

* Clearly mention to ignore negative sign. (paragraph given)

